Complete papers and shared theories

Nineteen topics, forty-two original texts

This candidate preserves all eighteen topic records from the admitted forty-one-original overview. It adds Mixes (F005), using its distinctly accepted complete numerical boundary. This consolidated overview still requires independent acceptance. Earlier editions remain preserved, with committed baseline and new packet hashes bound directly.

For each topic, $S = \sum_i c(L_i)$ counts its complete original texts, U every complete common paper, and $A = \sum_i c(A_i)$ every complete application. The comparisons are

$$R = \frac{S}{U + A}, \qquad r_i = \frac{c(L_i)}{c(A_i)} \quad \text{after acquiring the whole theory and declared software/data.}$$

Each topic pays its common theory independently, even when proofs overlap another topic. A factor above one means a shorter replacement in that coordinate; a factor below one means a longer one. Conditional factors do not include acquisition a second time, and are not standalone application savings.

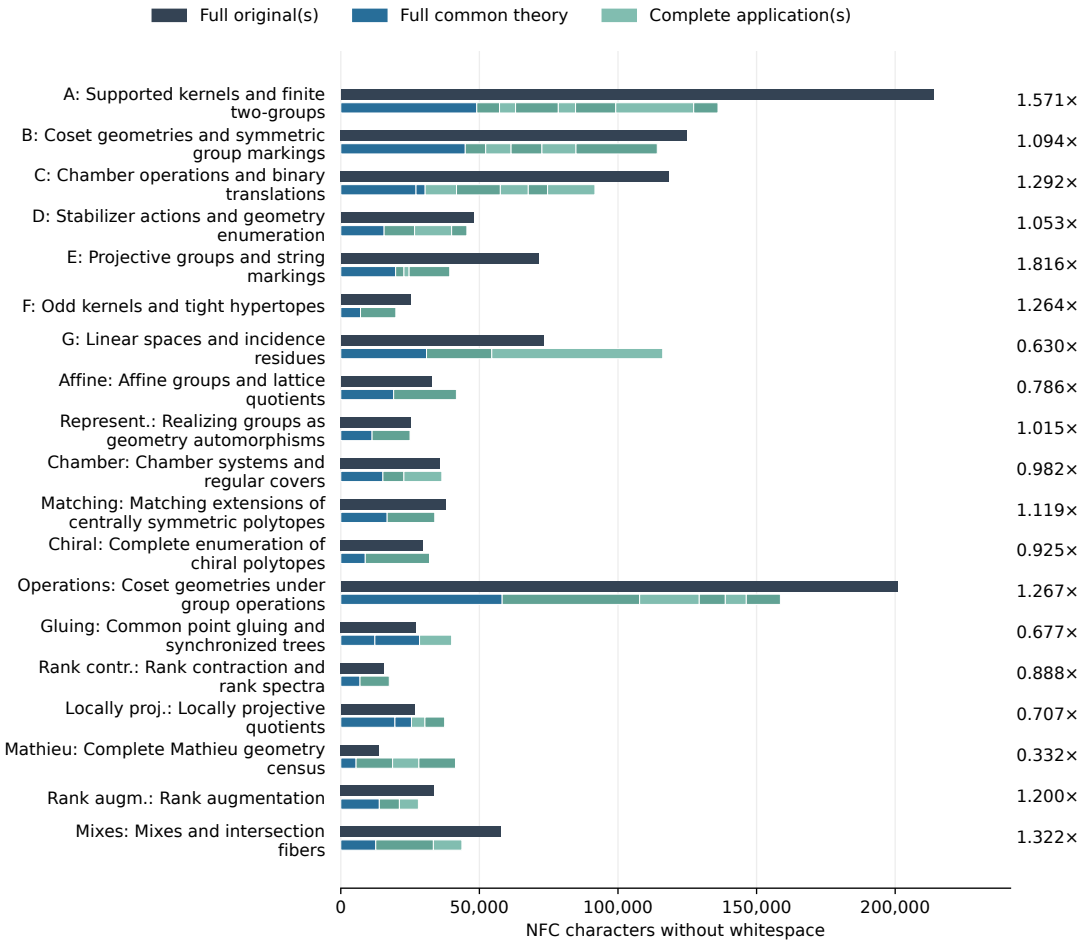
Complete written costs. All rows use NFC characters without whitespace at their stated representations. Full-document boundaries include introductions, context, proofs and references.

Topic	Originals	Sources	Common	Applications	Written	Factor
A	7	213,803	49,059	87,019	136,078	1.571
B	5	124,906	44,905	69,265	114,170	1.094
C	5	118,433	30,437	61,248	91,685	1.292
D	3	47,894	15,632	29,855	45,487	1.053
E	3	71,375	19,790	19,513	39,303	1.816
F	1	25,115	7,118	12,744	19,862	1.264
G	2	73,199	30,928	85,207	116,135	0.630
Affine	1	32,767	19,070	22,638	41,708	0.786
Represent.	1	25,348	11,288	13,688	24,976	1.015
Chamber	2	35,802	15,199	21,254	36,453	0.982
Matching	1	37,951	16,732	17,168	33,900	1.119
Chiral	1	29,599	8,831	23,171	32,002	0.925
Operations	3	200,891	58,215	100,379	158,594	1.267
Gluing	1	27,081	28,416	11,565	39,981	0.677
Rank contr.	1	15,530	6,937	10,555	17,492	0.888
Locally proj.	2	26,490	25,498	11,966	37,464	0.707
Mathieu	1	13,736	5,516	35,862	41,378	0.332
Rank augm.	1	33,604	13,904	14,096	28,000	1.200
Mixes	1	57,753	12,615	31,077	43,692	1.322

Representations remain distinct. B and C use complete authored-text renderings for N001 and L108, with raw untrimmed PDFs for their other documents. Those originals' unavailable-PDF costs remain unknown. Rank contraction, Rank augmentation, Mixes and the Mathieu census also use complete raw untrimmed PDFs. Locally projective and other rows use their accepted selected, calibrated PDF coordinate. Operations' encoded-event graphical proxies retain raw and text-only sensitivities. No pooled factor is formed.

Scope. The active catalogue contains 106 accessible originals: 42 admitted and measured, with 64 pending recovery. The 55 unavailable originals from the historical 161 are archived; retrieval is outside the current scope. Singleton topics remain included. These counts come from a separate root scope/admission snapshot, not lengths. Mixes has written factor 1.322; its program-inclusive factors 0.967 and 0.991 are below one. This overview adds no scientific, source, question or formal credit.

Complete common theories and applications. Dark bars are whole original sums. Blue segments are whole common papers; green segments are all complete applications. Unified B pays its entire current common paper once. C still pays its complete additional incidence note. The N006 application retains all three companions. Locally projective common theory includes both the proof and its complete tables; the Mathieu application includes the main paper, all class tables and all source-row dispositions. Rank augmentation pays its full common paper, application and counterproof. Mixes pays its whole common theory and both local papers.



All earlier common and application boundaries remain unchanged. Locally projective pays 7+5 common pages and both applications (2+3 pages). Mathieu pays its two-page common paper and all 5+10+9 application pages. Rank augmentation pays 5 common and 3+3 local pages; neither old B nor the earlier counterproof is free. Companion papers are fully charged. Mixes pays 5 common and 8+4 local pages. Text length is not minimum proof complexity or originality.

Complete programs and prior acquisition. The written comparison remains first. These alternatives add the entire declared code representation and any separately measured mandatory data. Raw and printed code are never added together. The data column is separate from code in both denominators.

Topic	Printed code	Data	With printed	Raw code	With raw	Upfront
A	54,210	–	1.124	52,807	1.132	103,269
B	13,170	–	0.981	12,330	0.987	58,075
C	20,933	–	1.052	19,937	1.061	51,370
D	27,070	–	0.660	26,185	0.668	42,702
E	7,565	–	1.523	–	–	27,355
F	8,837	–	0.875	8,061	0.899	15,955
G	53,018	–	0.433	50,184	0.440	83,946
Affine	17,250	–	0.556	16,682	0.561	36,320
Represent.	9,025	–	0.746	8,310	0.762	20,313
Chamber	7,677	–	0.811	6,401	0.835	22,876
Matching	4,998	–	0.976	4,408	0.991	21,730
Chiral	11,940	–	0.674	11,090	0.687	20,771
Operations	–	–	–	403,848	0.357	462,063
Gluing	13,371	–	0.508	12,768	0.513	41,787
Rank contr.	2,925	–	0.761	2,501	0.777	9,862
Locally proj.	22,695	28,574	0.299	22,104	0.301	76,767
Mathieu	31,244	48,686	0.113	30,489	0.114	85,446
Rank augm.	10,866	–	0.865	9,998	0.884	24,770
Mixes	16,025	–	0.967	14,608	0.991	28,640

Upfront means the whole common theory, complete printed code when available, and separately measured mandatory data. A dash in the Data column for an older record does not mean its external data is free; those historical boundaries remain unchanged. Locally projective adds all 28,574 data characters (39,150 literal bytes), and Mathieu all 48,686 (83,965 bytes), to both code alternatives. For Operations it uses all twenty-three whole mathematical/validation files plus both required decoding drivers. Its complete printed route is unavailable: the two partial appendices cover only four named raw files. Operations conditionals also assume acquisition of all required external data, separately inventoried as 21,960,689 representation bytes and 2,516,249 listed installed bytes. Byte storage is not added to character units. Large witnesses are not discounted.

E retains its separate optional-validation sensitivity: another 5,173 characters gives a program-inclusive factor of 1.372. No unadmitted raw-code alternative is invented for E.

Roles. A pays five files; B a finite premise, two validations and an auxiliary example. C pays three GAP files, including its exhaustive census and the shared construction once. D pays five required and three diagnostic programs. E separates required code from optional validation. F's three and Representation's two programs are validations. G pays four finite computations, two certificate bridges and seven validations; Affine six mathematical programs; Chamber one required and one diagnostic; Matching its census; Chiral all three files. Operations pays seventeen proof/witness files, six validations and both required decoders. Gluing pays three validations; its three drivers and two partial Lean files stay separate. Rank contraction pays its finite-example program, with driver and four partial declarations separate. Locally projective requires six programs and its full historical JSON; Mathieu seven programs and both JSON inputs. Their three drivers each and eleven/ten outputs are separate; neither has Lean declarations. Rank augmentation pays three programs: required finite examples, bounded validation and supporting counterexample check. Three drivers and two Lean files (15 scoped declarations) stay separate; no full source formalization is claimed.

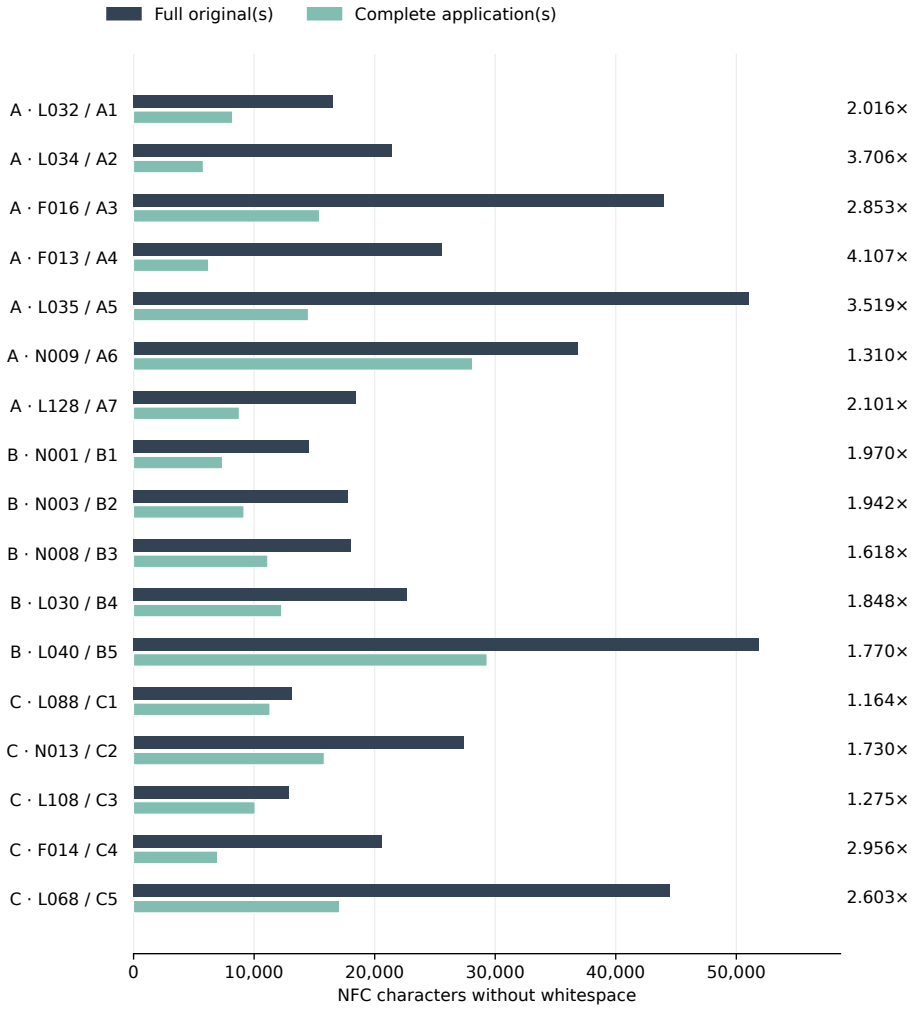
Limits. Complete external proofs, unprovided historical software, full graphics and end-to-end formalization have unknown costs, not zero. Every topic retains its underlying multi-unit and source-cover sensitivities. B's source costs are unchanged from the prior edition. This table does not compare a theorem closure with a complete original or use a node-count ratio.

Every complete application after full acquisition. The entire common theory and all declared software/data have already been acquired. Every local page is charged, including context, references and companions. DG2 includes all N006 papers; MC1 includes all 24 Mathieu application pages; RA1 includes both the application and counterproof; MX1 includes both Mixes papers.

Topic	Original	Application	Source cost	Application cost	Conditional
A	L032	A1	16,558	8,212	2.016
A	L034	A2	21,433	5,784	3.706
A	F016	A3	43,983	15,419	2.853
A	F013	A4	25,562	6,224	4.107
A	L035	A5	51,019	14,498	3.519
A	N009	A6	36,824	28,114	1.310
A	L128	A7	18,424	8,768	2.101
B	N001	B1	14,519	7,371	1.970
B	N003	B2	17,780	9,155	1.942
B	N008	B3	18,009	11,132	1.618
B	L030	B4	22,704	12,286	1.848
B	L040	B5	51,894	29,321	1.770
C	L088	C1	13,166	11,311	1.164
C	N013	C2	27,361	15,815	1.730
C	L108	C3	12,849	10,075	1.275
C	F014	C4	20,577	6,962	2.956
C	L068	C5	44,480	17,085	2.603
D	L066	D1	20,916	10,989	1.903
D	L074	D2	20,661	13,367	1.546
D	L104	D3	6,317	5,499	1.149
E	L083	E1	20,885	3,002	6.957
E	L079	E2	20,908	1,855	11.271
E	L021	E3	29,582	14,656	2.018
F	L014	F1	25,115	12,744	1.971
G	L009	G1	49,891	23,544	2.119
G	L092	G2	23,308	61,663	0.378
Affine	L012	AF1	32,767	22,638	1.447
Represent.	F010	R1	25,348	13,688	1.852
Chamber	L008	H1	21,784	7,571	2.877
Chamber	N016	H2	14,018	13,683	1.024
Matching	F015	M1	37,951	17,168	2.211
Chiral	L028	CE1	29,599	23,171	1.277
Operations	N005	OP1	122,002	49,595	2.460
Operations	N006	DG2	56,001	38,479	1.455
Operations	N007	DG1	22,888	12,305	1.860
Gluing	L010	GL1	27,081	11,565	2.342
Rank contr.	L037	RC1	15,530	10,555	1.471
Locally proj.	L084	LP1	2,071	4,857	0.426
Locally proj.	L095	LP2	24,419	7,109	3.435
Mathieu	L073	MC1	13,736	35,862	0.383
Rank augm.	L020	RA1	33,604	14,096	2.384
Mixes	F005	MX1	57,753	31,077	1.858

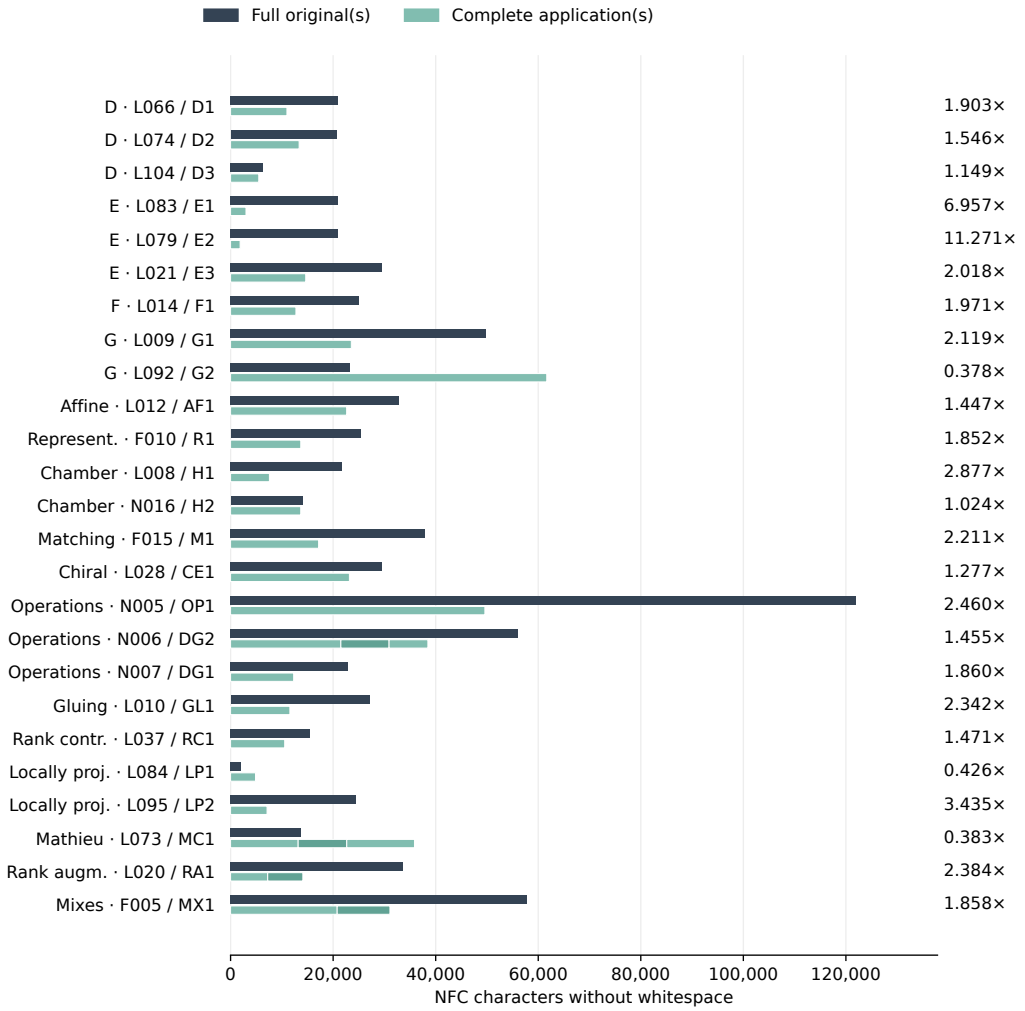
All eighteen previous topic records remain unchanged. One complete application is added. Recovery counts come from the separate admission snapshot; no scientific credit is assigned here.

Individual source and application costs: first seventeen originals. Dark bars show complete original texts; green bars show complete applications. Factors at right assume full prior acquisition. The source and replacement bars share the same scale within each chart.



Source IDs remain stable catalogue cross-references. LP1 is the complete 2006 erratum; LP2 uses the full fifteen-page 2003 arXiv edition, not the unseen journal text. Their source-column corrections remain qualified. MC1 gives the corrected 81/431-class census and all 431 literal-row dispositions, not unavailable historical `geo[]` identities. Its 48-page predecessor is retained notation, not a newly recovered original. L020 cases 1 and 2 keep their hypotheses; false case 3 is replaced by the corrected criterion and full S18 counterproof. Its motivating Table 1 census is retained, not regenerated; higher-rank questions remain open.

Individual source and application costs: remaining twenty-five originals. The same conventions apply. LP1 and LP2 assume acquisition of the whole Locally projective proof, common tables, all six programs and mandatory JSON. MC1 contains all three Mathieu application papers, after whole common theory, all seven programs and both JSON inputs. RA1 pays both local papers after its full five-page common theory and three programs. MX1 pays all 8+4 local pages after acquiring its whole five-page common paper and all three programs, printed (4+3 pages) or raw. Other topics remain separately paid; no overlap is deducted.



Mixes retains 35 principal/background dispositions: five literal counterexample targets and two convention-qualified targets, with corrected criteria. All three final problems remain unanswered. Its programs are one required finite certificate and two validations; three drivers and one Lean file with two generic declarations stay separate.

Data and provenance. Exact integers and rational comparisons are in `results.json`, `topics.csv` and `applications.csv`. The protocol pins committed unchanged numerical and admission records and the new frozen numerical packet and its distinct root acceptance directly. No primary text is copied or extracted. Bar segments use complete document costs, not theorem counts. Distinct review is required before this overview becomes the public current edition.